

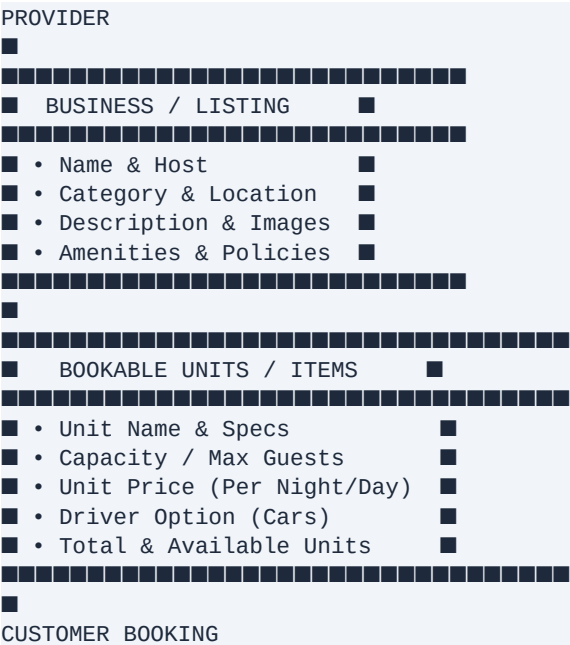
EASYSERVICE MARKETPLACE — UNIFIED BOOKING ARCHITECTURE & DOMAIN MODEL SPECIFICATION

Course: Software Testing and Quality Assurance / Software Architecture
Institution: Addis Ababa University — School of Information Technology and Engineering
Document Type: Domain Model Specification & Unified Booking Engine Architecture

1. CENTRAL ARCHITECTURE PHILOSOPHY

> "A provider owns a business/listing, and that listing contains bookable units, options, or inventory."

EasyService enforces a unified domain model where a single booking engine operates across four primary service categories (**Hotels & Stays, Drive & Car Rentals, Experiences & Events, Shop & Authentic Crafts**) without duplicating backend services.



2. UNIFIED DOMAIN MODELS

2.1 Provider Business Model (`BusinessListing`)

```
{
  "id": "prov_kuriftu_1",
  "name": "Kuriftu Resort & Spa Bishoftu",
  "category": "HOTEL",
  "location": "Bishoftu Lakefront, Oromia",
  "hostName": "Kuriftu Hospitality Group",
}
```

```
"description": "Serene luxury lakefront resort offering spa wellness, fine dining, private pool villas, and lush gardens.",
"rating": 4.8,
"reviewCount": 240,
"amenities": ["Free WiFi", "Swimming Pool", "Spa & Wellness", "Lakefront Dining", "Airport Shuttle"],
"policies": {
  "checkInTime": "14:00",
  "checkOutTime": "11:00",
  "cancellation": "Free cancellation up to 24h before check-in"
}
```

2.2 Category Bookable Units Specifications

A. ■ HOTEL (Stays & Accommodation)

```
{
  "unitId": "unit_k1",
  "name": "Standard Room",
  "maxGuests": 2,
  "bedType": "1 Double Bed",
  "roomSize": "28 m²",
  "pricePerNight": 3500,
  "availableUnits": 8,
  "amenities": ["WiFi", "Breakfast Included", "Garden View"]
},
{
  "unitId": "unit_k2",
  "name": "Deluxe King Room",
  "maxGuests": 2,
  "bedType": "1 King Bed",
  "roomSize": "35 m²",
  "pricePerNight": 5000,
  "availableUnits": 4,
  "amenities": ["WiFi", "Jacuzzi", "Lake View", "Breakfast Included"]
},
{
  "unitId": "unit_k3",
  "name": "Family Villa Suite",
  "maxGuests": 5,
  "bedType": "2 Bedrooms (1 King + 2 Singles)",
  "roomSize": "85 m²",
  "pricePerNight": 10000,
  "availableUnits": 2,
  "amenities": ["Private Pool", "Full Kitchen", "Butler Service"]
}
```

B. ■ CAR RENTAL (Drive & Vehicle Fleets)

Includes the **Driver Choice Option** (`withDriver: true (+500 ETB/day)` or `selfDrive: true`):

```
{
  "unitId": "unit_c1",
  "name": "Toyota Land Cruiser V8 4x4",
  "year": 2024,
  "transmission": "Automatic",
  "seats": 7,
  "fuelType": "Diesel",
  "pricePerDay": 3500,

```

```

"driverFeePerDay": 500,
"driverOptionAvailable": true,
"availableUnits": 3,
"badge": "4x4 Off-Road"
},
{
"unitId": "unit_c2",
"name": "Hyundai Tucson SUV",
"year": 2023,
"transmission": "Automatic",
"seats": 5,
"fuelType": "Petrol",
"pricePerDay": 2500,
"driverFeePerDay": 500,
"driverOptionAvailable": true,
"availableUnits": 5,
"badge": "City SUV"
}

```

C. ■ EXPERIENCES & EVENTS (Ticket Tiers)

```

{
"unitId": "unit_e1",
"name": "Regular Entrance Pass",
"price": 500,
"availableQuantity": 150,
"perks": ["Main Stage Access", "Food Court Area"]
},
{
"unitId": "unit_e2",
"name": "VIP Elevated Pass",
"price": 1500,
"availableQuantity": 30,
"perks": ["VIP Lounge", "Free Drinks", "Elevated Stage View"]
},
{
"unitId": "unit_e3",
"name": "VVIP Artist Meet & Greet Pass",
"price": 3000,
"availableQuantity": 10,
"perks": ["Backstage Access", "Artist Meet & Greet", "Dinner Buffet"]
}

```

D. ■ SHOP & CRAFTS (Inventory Products)

```

{
"unitId": "unit_s1",
"name": "Authentic Ceramic Jebena Set",
"price": 450,
"stockQuantity": 20,
"options": ["Standard Clay", "Decorated Gold Border"]
},
{
"unitId": "unit_s2",
"name": "Handwoven Silk Habesha Kemis",
"price": 3200,
"stockQuantity": 8,
"options": ["Size M", "Size L", "Size XL"]
}

```

3. UNIFIED PRICING ENGINE CALCULATIONS

The core **PricingService** executes category-specific subtotal logic:

$$\text{Hotel Subtotal} = \text{pricePerNight} \times \text{nights} \times \text{rooms}$$

$$\text{Car Subtotal} = (\text{pricePerDay} + (\text{hasDriver} ? 500 : 0)) \times \text{days} \times \text{vehicles}$$

$$\text{Event Subtotal} = \text{ticketPrice} \times \text{tickets}$$

$$\text{Store Subtotal} = \text{productPrice} \times \text{quantity}$$

$$\text{Final Total} = \text{Subtotal} - \text{Discount} + \text{Tax (15\% VAT)}$$

4. BOUNDARY VALUE ANALYSIS (BVA) TEST MATRICES

Category	Test Case Parameter	Boundary Values	Expected Result	System Action
Hotel	Stay Duration (Nights)	`0 nights`	■ Invalid	Error: Minimum stay is 1 night
Hotel	Stay Duration (Nights)	`1 night`, `2 nights`, `30 nights`	■ Valid	Calculates subtotal correctly
Hotel	Room Stock Depletion	`Booking N = Stock`	■ Valid	Units remaining becomes 0 (Sold Out)
Hotel	Overbooking Guard	`Booking N > Stock`	■ Blocked	Reject transaction with Concurrency Lock
Car Rental	Driver Selection	`Self-Drive` vs `With Driver (+500)`	■ Valid	Adds 500 ETB/day fee to subtotal
Car Rental	Dates Validity	`Return Date < Pickup Date`	■ Invalid	Error: Invalid date range
Event	Ticket Order	`Quantity = 0`	■ Invalid	Error: Minimum 1 ticket required
Event	Ticket Order	`Quantity = Remaining Capacity`	■ Valid	Ticket stock successfully decremented

5. SUMMARY

This unified architecture guarantees that EasyService scales across diverse service domains with a single, highly testable backend booking engine and a consistent customer user interface.